

Discharge Summary

Patient: Brigitte Schuster
Date of Birth: 10/09/1959
Admission Date: 07/07/2025
Discharge Date: 07/15/2025
Attending Physician: Dr. Christine Weber, MD

Dear Colleagues,

Ms. Brigitte Schuster, DOB 10/09/1959, was admitted on 07/07/2025 in diabetic ketoacidosis. She has had type 1 diabetes since age 19 and manages with a continuous subcutaneous insulin infusion pump. She developed nausea, vomiting, polyuria, and fatigue over 2 days following a gastroenteritis illness during which she had substantially reduced her prandial insulin boluses.

Primary Diagnosis: Type 1 diabetes mellitus with ketoacidosis (E10.1)

Laboratory values on admission and at DKA resolution:

Parameter	Value	Reference
HbA1c	12.1%	< 7.0%
pH (admission)	7.18	7.35–7.45
pH (resolution)	7.36	7.35–7.45
Glucose (admission)	498 mg/dL	70–100 mg/dL

Past Medical History: Type 1 diabetes for 35 years. Mild background diabetic retinopathy — last reviewed 14 months ago, no treatment required. Microalbuminuria (UACR 48 mg/g), stable eGFR above 90 mL/min. Coeliac disease diagnosed 2004 — strict gluten-free diet. Hypothyroidism on levothyroxine 100 mcg. Non-smoker, no alcohol. Part-time physiotherapist.

DKA Management: IV insulin infusion at 0.1 units/kg/hour. IV fluid resuscitation: 1 L over 60 min initially, then adjusted to clinical assessment and urine output. Potassium replacement once urine output confirmed.

Arterial blood gas, blood ketones, and electrolytes monitored hourly for 6 hours, then 2-hourly. DKA resolution criteria met at 16 hours (pH 7.36, bicarbonate 22 mmol/L, blood ketones 0.4 mmol/L). Insulin pump restarted day 2 with revised basal profile.

Root Cause Analysis: Diabetes technology team downloaded pump data on day 3. A kinked cannula in the previous insertion site was identified as the likely precipitant for relative insulin deficiency. Guidance on cannula site rotation, insertion technique, and early recognition of pump occlusion was reinforced.

Sick Day Rules Education: Structured session by diabetes specialist nurse day 3 — 'never stop insulin' principle, ketone monitoring during illness, prandial and basal rate adjustments during gastroenteritis, when to seek emergency care. Ms. Schuster demonstrated excellent understanding and was given a written sick day rules card for her wallet.

Psychological Impact: Psychological assessment by ward psychologist day 5. No clinical anxiety or depression on standardised screening. One-session intervention normalising the DKA experience and reinforcing pump self-management efficacy delivered. Contact details for diabetes psychology outpatient service provided.

Levothyroxine reviewed — TFTs showed adequate replacement, no change required.

Medication at Discharge: Insulin pump with revised basal profile, levothyroxine 100 mcg, omeprazole 20 mg.

Follow-up: Diabetes outpatient in 2 weeks with pump data download and HbA1c. Ophthalmology in 3 months. Nephrology in 6 months. GP in 1 week.

DKA Protocol — Detailed Management Log:

On arrival, Ms. Schuster's parameters confirmed severe DKA: pH 7.18, bicarbonate 8 mmol/L, blood ketones 4.2 mmol/L, glucose 498 mg/dL. IV insulin at 0.1 units/kg/hour was commenced immediately. IV 0.9% NaCl: 1 L over 60 minutes, then 500 mL/hour for 2 hours, then adjusted to clinical status

and urine output. Potassium replacement was initiated once urine output confirmed and potassium fell to 5.2 mmol/L. Hourly monitoring of blood gas, blood ketones, glucose, and electrolytes was performed for the first 6 hours, then 2-hourly thereafter. DKA resolution criteria met at hour 16. Transition from IV to subcutaneous pump insulin was achieved on day 2.

Coeliac Disease and Nutritional Considerations:

Ms. Schuster has confirmed coeliac disease and maintains a strict gluten-free diet. During the acute DKA phase, oral intake was not possible and nutritional requirements were met via IV fluids. As oral intake resumed on day 2, the ward catering team was briefed and all meals provided were confirmed gluten-free. The dietitian reviewed on day 3 — nutritional status was satisfactory on her established diet. No supplemental feeding was required. The dietitian also reviewed carbohydrate intake management during gastroenteritis-related appetite loss and provided updated guidance on insulin dose adjustment without omitting insulin.

Psychological Support and Self-Efficacy:

Psychological assessment on day 5 showed no clinical anxiety or depression on standardised screening. A one-session intervention was delivered by the ward psychologist normalising the DKA experience and reinforcing self-efficacy in pump management. Ms. Schuster expressed significant distress about the episode and concern about her ongoing safety — these concerns were addressed empathically while reinforcing her demonstrable competence in pump therapy. Contact details for the diabetes psychology outpatient service were provided and she was encouraged to self-refer if ongoing support would be beneficial.

Dr. Christine Weber, MD

Specialist in Endocrinology